

AUSNVIRO INFORMATION BOOKLET - INTRODUCING THE NEW 7 STAR REQUIREMENT.



Introducing 7 Star & Whole of Home ratings

The 2022 National Construction Code (NCC) requires a new minimum thermal performance rating of 7 stars under the Nationwide House Energy Rating Scheme (NatHERS). In addition, all new homes must also meet a minimum annual energy use budget.

What's changing?

The updated efficiency requirements for new residential buildings (detached houses, townhouses and apartments) included in the 2022 edition of the National Construction Code cover:

- An increase in the minimum efficiency performance requirement for the building shell (insulation, window glazing etc) from 6 to 7 stars out of a possible 10 through the Nationwide House Energy Rating Scheme (NatHERS)
- Introduction of a Whole-of-Home annual energy use budget for fixed appliances (hot water, heating and cooling, lighting and pool and spa pumps) with potential to offset with rooftop solar
- Removal of the current Victorian variation that requires either a solar water heater or a rainwater tank to be installed, allowing Victorians to choose their preferred hot water system
- Requirements to ensure apartments are designed for easy and cost-effective retrofit of solar systems and electric vehicle charging.

What do these changes mean in practice?

Getting to 7 stars

In most cases, the new standards can be met through relatively simple and low-cost changes, particularly if design is considered early in the process of a new home build. Key differences between a 6 star and 7 star home may include:

- Better design and orientation to the sun to take advantage of opportunities to passively heat the home in winter while shading it in summer
- Improved ceiling, wall or under-floor insulation to suit your climate
- High performance glazing (windows) to better insulate the home from heat loss in winter and heat gain in summer.

During the early roll out of these new standards, some volume builders may need to adjust their standard designs to 7 star rating until they become commonplace. You may find that a design you like needs minor modifications to help you achieve the minimum 7 star rating.

What does this mean for our clients ?

The time is approaching with the new 7 star requirement being made mandatory as of the 1st of October 2023.

Ausnviro are committed to assisting all our existing and new clients to navigate through this process. We have put together this information booklet which compiles some of the industries leading information and making it readily available.

Our aim is not to go into 'information overload' but provide you with an accessible source of relevant documentation and information that hopefully assists you in this transition.

It should be noted that majority of the dwellings we assess would not achieve 7 stars even with major insulation and glazing upgrades. This is also the consensus throughout the industry and has become very apparent through topic discussion and surveys.

How can we help you ?

We are strongly suggesting all new projects be sent through to us, for a preliminary assessment to determine the thermal performance at concept stage. This can help us find any major issues before it's too late and provide valuable feedback to not only reduce upgrade costs but ensure the proposed design can achieve the minimum compliance.

Preliminary assessments will be quoted per dwelling, but this will be a minor cost to your clients and will provide the necessary feedback required and could save them thousands of \$\$ and also the disappointment of major design changes late in the process at BA approval stage.

“We are strongly suggesting all new projects be sent through to us for a preliminary assessment to determine the thermal performance at concept stage.”



Whole of Home (WOH) ?

Included alongside the existing thermal star rating, the new NatHERS Whole of Home rating considers the energy used by heating, cooling, cooking and plug in appliances; as well as hot water systems, lighting, and pool and spa equipment. The rating can demonstrate how a new home meets the NCC's new energy use budget.

The 0-100 rating includes the energy used for heating, cooling, and appliances minus energy generated via solar panels. This provides a useful snapshot of costs as well as greenhouse gas emissions. You can check out [abcb.gov.au/news/2022/ncc-2022-here](https://www.abcb.gov.au/news/2022/ncc-2022-here) to learn more about local NCC 2022 compliance requirements in your state or territory.

What additional information is required to complete WOH?

In order to complete the new required WOH assessment and report the following additional information will need to be provided either on the architectural plans or the way of a specification document.

Heating and cooling appliances

- Heating and cooling appliance types (ideally including models and sizes)
- Appliance efficiencies (with metrics appropriate to the appliance types)
- Mark-up of NatHERS conditioned zones to be serviced by the heating and cooling appliances (several different areas in a home may be heated/cooled, and different equipment may be specified in each of these areas)

Hot water system

- Hot water service type (ideally including model and size)
- Appliance efficiency (with metric appropriate to the appliance type)

Lighting

- Calculated lighting power density (W/m²) complying with Section 13.7.6 of the ABCB Housing Provisions

Pool and spa pumps (if applicable)

- Pool dimensions (either the total surface area of the pool (m²) or the total volume of water (litres) that the pool is designed to hold)
- Pool pump model and type (single-, double- or multispeed)
- Pool cleaner type
- Pump efficiency expressed as a star rating (based on the energy rating label commencing 1 October 2022)

Renewable energy (if applicable)

- Rated sizes (in kilowatt – kW) of PV arrays – if there is more than one group of PV panels, then the size of each group is required
- Azimuth (orientation) and inclination (slope) of PV arrays – if there is more than one group of PV panels, then these details are required for each array location
- Rated inverter size (kW), number of phases (single-, two- or three-phase) and export limits (kW)

Please note if any of the above information is not available, defaults will be used and this could effect the compliance outcome.



New NatHERS Ratings

Expanding and improving energy ratings across Australia

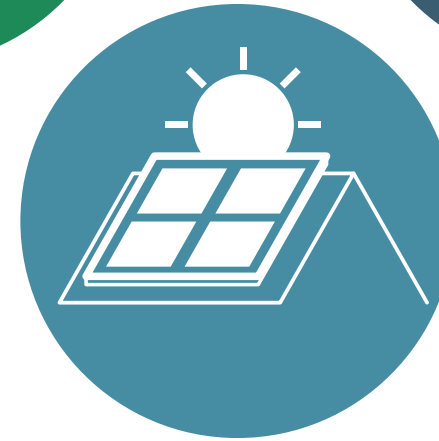
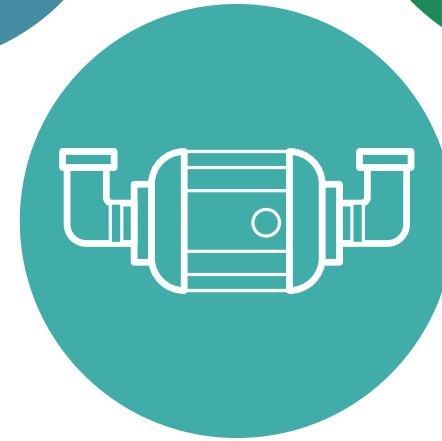
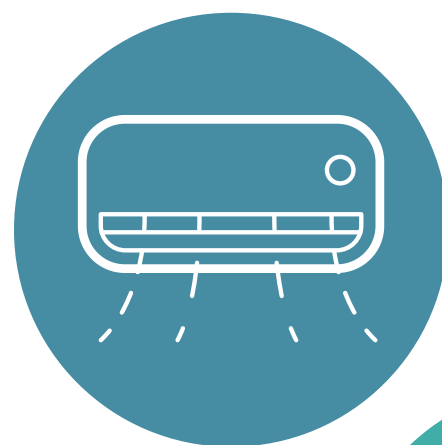
NatHERS is Expanding

NatHERS will rate the energy performance of your fixed appliances such as heating, cooling, hot water, solar panels and batteries. This is in addition to the thermal shell rating for your home.



Consumer Choice

A NatHERS assessment helps you improve the energy efficiency of your planned home or major renovation. It also helps you decide what appliances and energy sources best suit your needs.



Updated Certificate

New certificates will have two ratings: a star rating out of 10 and a 0-100 performance rating. Higher scores mean your home will be more comfortable to live in and cheaper to run.



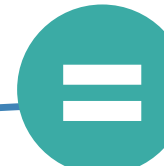
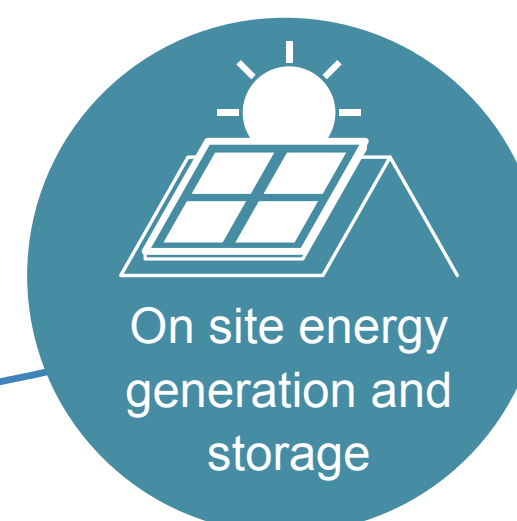
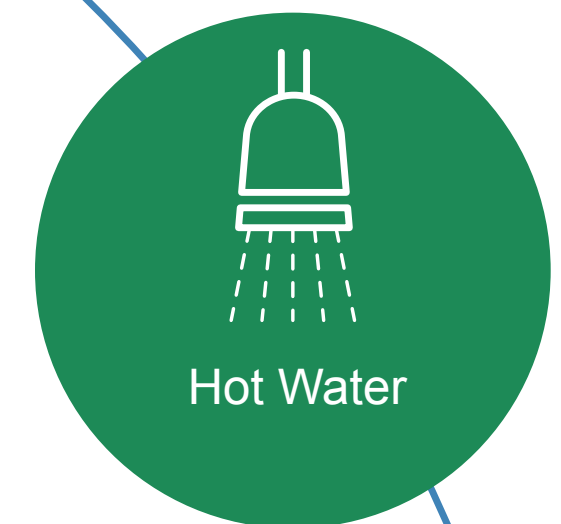
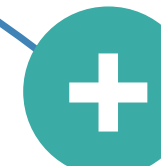
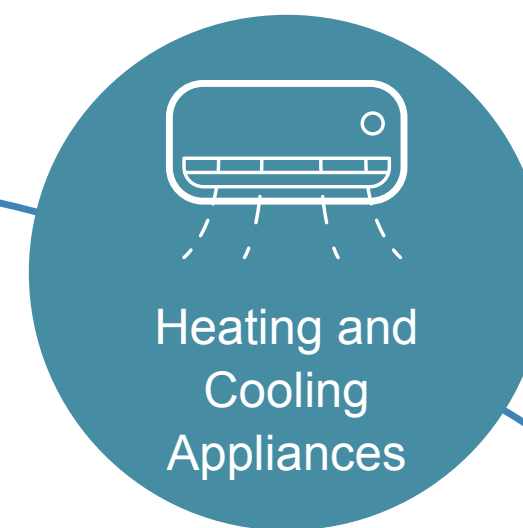
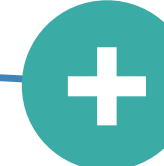
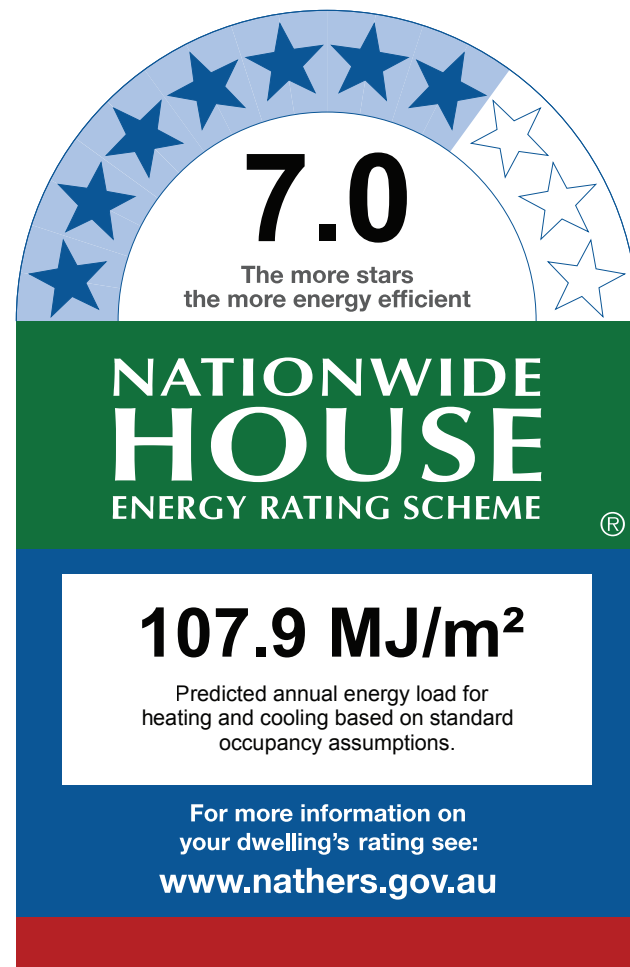
The Benefits



NatHERS helps you build a comfortable, energy efficient, healthy home for your family with lower energy bills and reduced emissions.



Using NatHERS to pass Energy Efficiency Requirements in the National Construction Code (NCC)



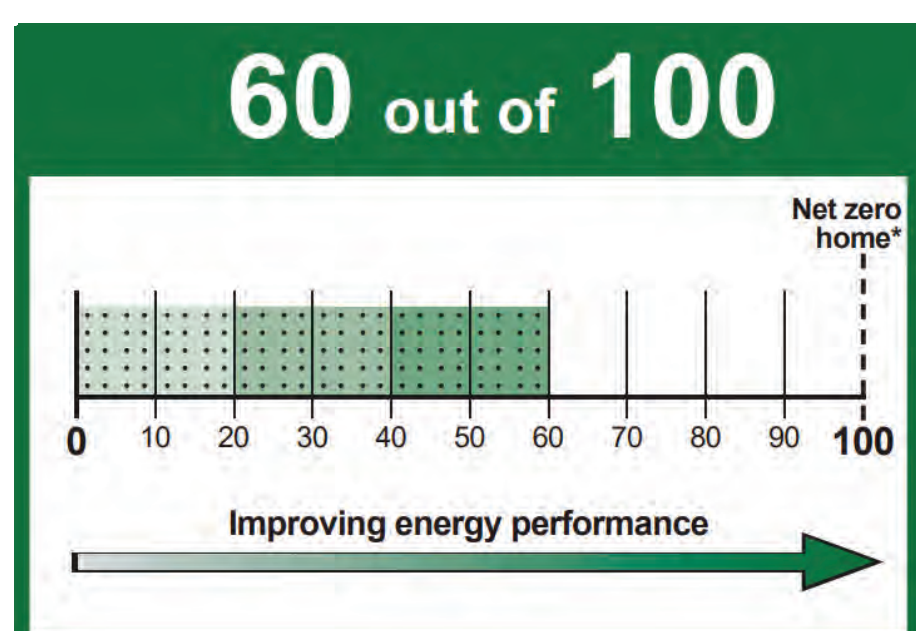
Thermal Star Rating

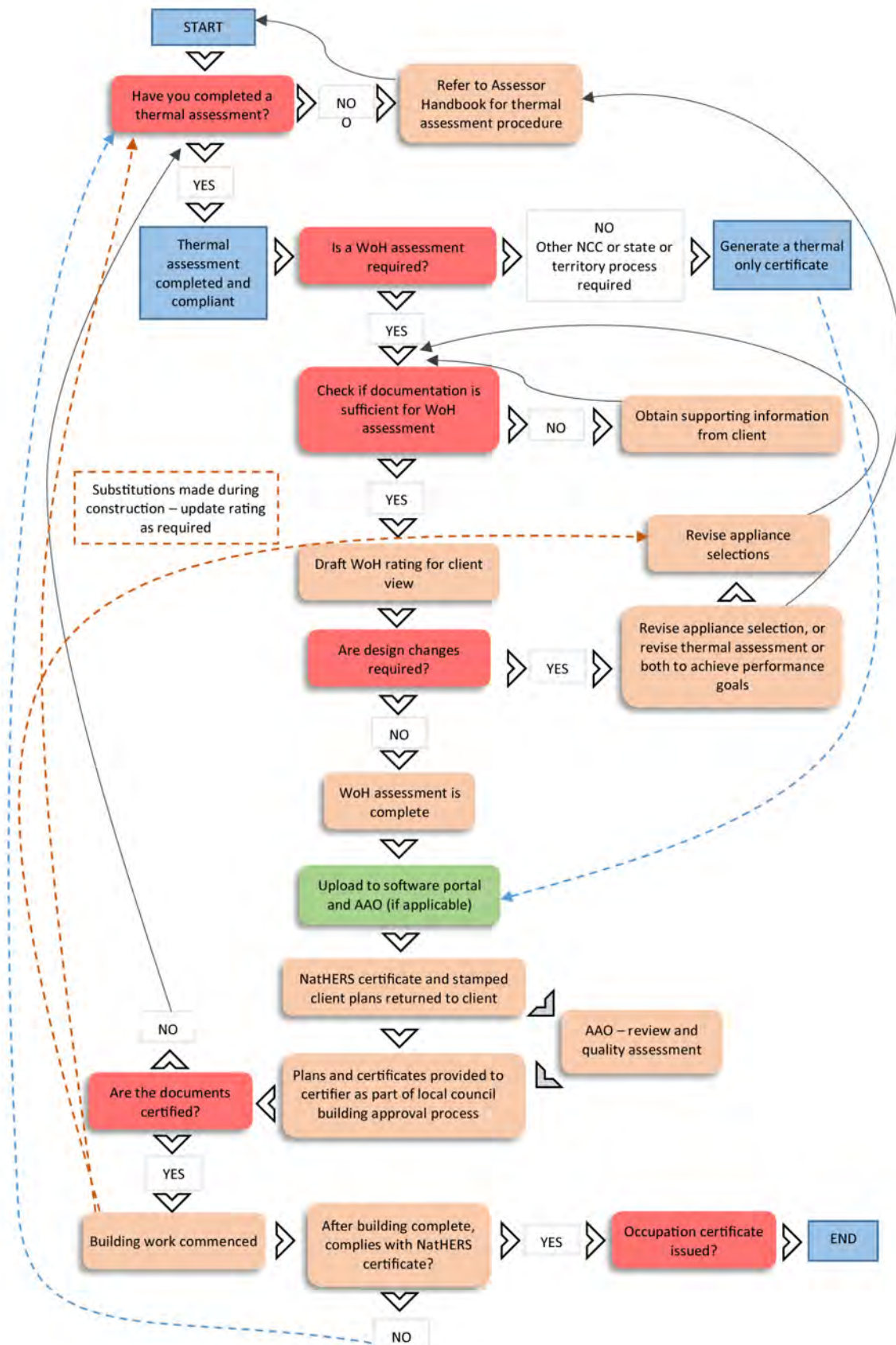
First achieve thermal rating at the minimum or above National Construction Code (NCC) requirement. A NatHERS thermal rating (stars out of 10) is based on information about the home's design, construction materials and the climate where it is being built and assesses the thermal performance (heating and cooling needs) of a home's construction.



Whole of Home Rating

It is a NCC requirement to meet or exceed the minimum Whole of Home ratings. Once your thermal rating is established you can add home appliances and onsite energy generation to generate your Whole of Home rating out of 100.





AAO = assessor accrediting organisation; NCC = National Construction Code; WoH = Whole of Home

Figure 1: NatHERS Whole of Home assessment



Australian Government

Top tips for building for 7 Stars

What's changed?

The 2022 National Construction Code (NCC) will require all new Australian houses and apartments to meet a minimum energy efficiency rating of 7 stars under the Nationwide House Energy Rating Scheme (NatHERS). Equivalent increases exist for other NCC compliance pathways: Deemed-to-Satisfy elemental provisions; and the Reference Building Verification Method. There will be allowances for 6 and 6.5 star houses with a compliant outdoor living area in NCC climate zones 1 and 2. See Section S42C2 of Volume Two for more details, <https://ncc.abcb.gov.au>.

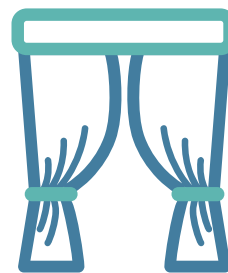
Getting your dwelling to 7 stars

Building for 7 stars doesn't have to be costly or complicated. Simple design features and material choices make a big difference to energy efficiency and the rating outcome.

- 1 Get your NatHERS assessor involved early:**
The most important first step is to get your NatHERS assessor involved early in the design process. NatHERS accredited assessors are extensively trained to improve ratings and providing options to get to 7 stars and beyond. Find a NatHERS accredited assessor here. www.nathers.gov.au/owners-and-builders/home-energy-star-ratings.

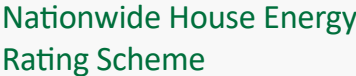
- 2 Orientation is key:** Build for the local climate, allowing shade and solar access as appropriate for tropical/temperate/cool climates. Repositioning a house plan on the block can make a world of difference to its performance. See <https://www.yourhome.gov.au/passive-design/orientation>

- 3 Improve the glazing:** The cost of double and triple glazing has reduced considerably in recent years and there are thousands of options now available. NatHERS includes both default and custom window options.



Choosing defaults will usually result in a lower rating, so switching to a custom window where all the specifications are known allows for greater flexibility. Window sizing, placement and shading are also important considerations and should suit the climate. See <https://www.yourhome.gov.au/passive-design/glazing> to find out more.





Changes to the 2022 NatHERS Certificates

New NatHERS Certificates have been developed to help industry and consumers design and build new energy efficient homes and to support the economy's transition to net zero emissions by 2050.

New Certificate Features

- NatHERS now includes a new Whole of Home performance rating that measures the energy use of the whole home including appliances, solar and batteries.
- The new Whole of Home rating can be used to show how the home meets or beats the energy efficiency requirements in the National Construction Code (NCC) 2022.
- Providing new information for the first time for households on what appliances have the biggest impact on their homes bills and carbon footprint.
- An expanded checklist for verifying that the home is designed and then built according to the information used to create the NatHERS ratings.
- New schedules for thermal bridging, appliances, onsite renewable energy and battery systems showing the home owner the details entered by the assessor to create the NatHERS ratings.

Thermal Performance Star Rating

The new NatHERS Certificate will retain the thermal performance star rating to demonstrate compliance with the thermal performance requirement in NCC 2022.

NCC Requirements

New information has been added about the Volume of the NCC the assessment has been done under and if a state or territory variation to the NCC applies.

The information on the NCC requirements has been updated to reflect the NCC 2022 energy efficiency provisions.

Sample of a house (class 1) and apartment (Class 2 and 4 individual unit) Certificate

Nationwide House Energy Rating Scheme®

NatHERS® Certificate No. #000000000-00

Generated on [date] using [software and version]

[other boilerplate text other boilerplate text other boilerplate text other boilerplate text other boilerplate text other boilerplate text]

Thermal performance star rating

7.0

The more stars
the more energy efficient

Property

Address [00 Street, Suburb, State/Territory, Postcode]

Lot/DP [number]

NCC class* [number]

Floor/all Floors [dwelling entrance floor] of [total no. of floors] floors

Type [new/renovation/existing]

Plans

Main plan [plan number, version & date]

Prepared by [name of preparer of plans]

Construction and environment

Assessed floor area (m²)*

Conditioned*	000.0
Unconditioned*	0.0
Total	0.0
Garage	0.0

Exposure type [exposure]

NatHERS climate zone [number, town/suburb]

Accredited assessor

Name [assessor name]

Business name [business name]

Email [email address]

Phone [00 0000 0000]

Accreditation No. [0000 000 000]

Assessor Accrediting Organisation [name of Assessor Accrediting Organisation]

Declaration of interest [declaration]

NCC Requirements

BCA provisions [Volume 1/Volume 2]

State/Territory variation [Yes/No]

National Construction Code (NCC) requirements

The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J3D3 and J3D15 of NCC Volume One.

NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.

The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au.

Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.

Thermal performance (MJ/m²)

Limits taken from ABCB Standard 2022.1

	Heating	Cooling
Modelled	0000.0	0000.0
Load limits	0000.0	0000.0

Features determining load limits

	[Type]
Floor type (lowest conditioned area)	
NCC climate zone 1 or 2	[Y/N/NA]
Outdoor living area	[Y/N/NA]
Outdoor living area ceiling fan	[Y/N/NA]

Whole of Home performance rating

60 out of 100

Net zero home*

Improving energy performance

Verification

To verify this certificate, scan the QR code or visit hstar-dev.azurewebsites.net/QR/Generate?p=MIaCPjgJ

When using either link, ensure you are visiting hstar-dev.azurewebsites.net

* Refer to glossary.
Generated on [date] using [software] for [address]

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For further information please visit: www.nathers.gov.au

Thermal Performance

New information about the heating and cooling load limits and features determining the load limits has been added to the Certificate. This information will allow users of the Certificate, such as certifiers, to quickly assess whether the NCC load limits have been met, while also allowing them to consider local variations. Additional explanatory information about the limits and settings is provided on page 2 of the Certificate.

Whole of Home performance rating

This new Whole of Home rating scale has been developed to show how the home meets or beats the new NCC annual energy use budget. This will take into consideration the efficiency of the appliances used in a new home including:

- heating and cooling
- hot water systems
- lighting
- pool/spa equipment
- cooking and plug-in appliances
- on-site energy generation and storage

The Whole of Home Performance Rating will be a separate rating out of 100. The new assessment builds on the thermal performance assessment rating and considers energy used for heating and cooling, and appliances, minus energy generated from solar panels, building a useful energy snapshot of energy costs as well as greenhouse gas emissions.

While the Whole of Home rating scale ranges from 0 to 100 (where 100 is a net zero energy value home), ratings above 100 are possible. One way that a home may rate over 100 is where a home generates more energy than it uses.

Additional information provided on page 2 of the Certificate can be used to indicate if the home is 'net zero energy' or 'net zero carbon.'



Nationwide House Energy Rating Scheme

Changes to the 2022 NatHERS Certificates

About the ratings

New explanatory information is provided about each of the NatHERS ratings. This information is designed to ensure users of the Certificate understand what each rating is measuring.

Heating and Cooling Load Limits

New explanatory information is provided about the heating and cooling load limits.

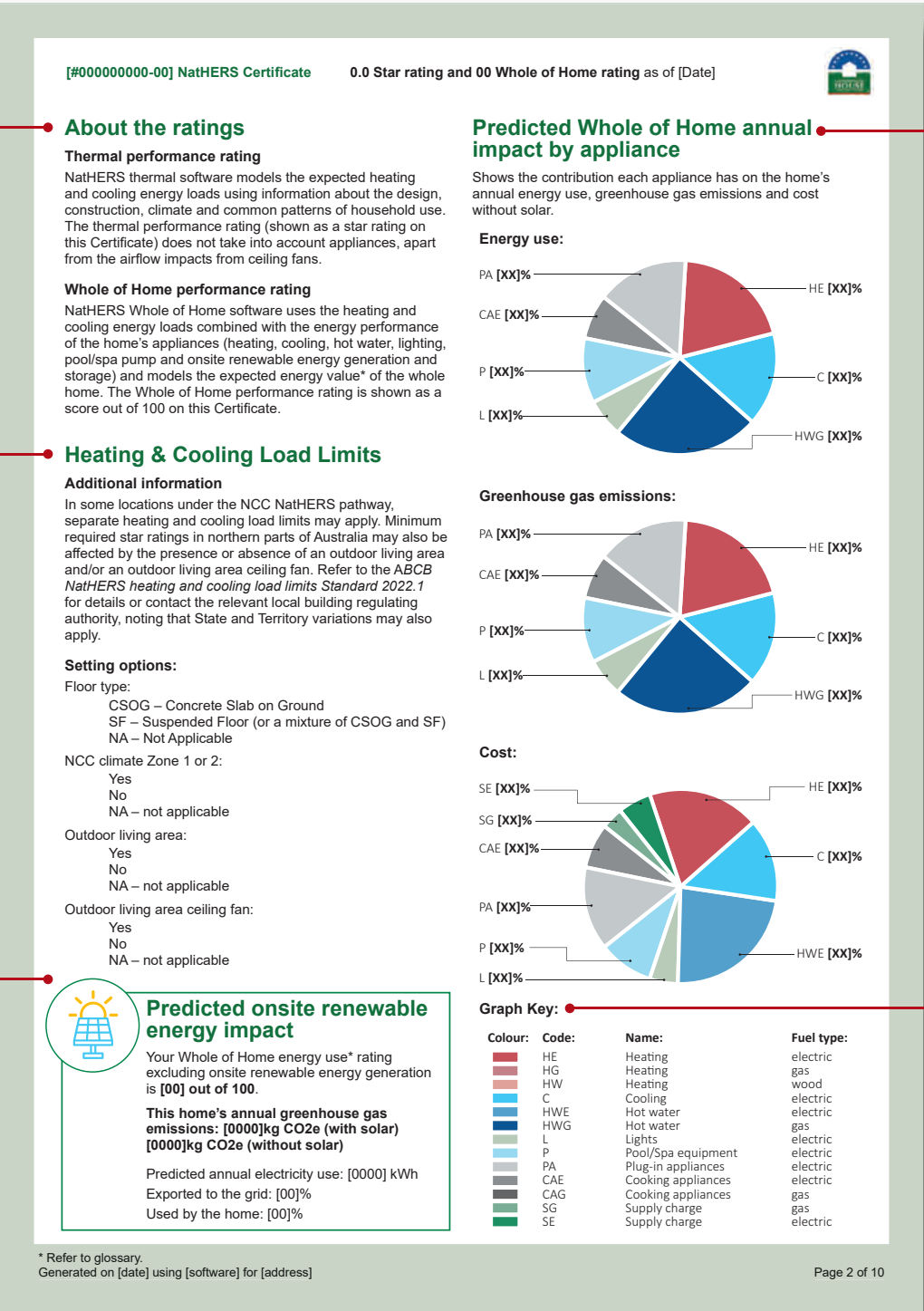
This information is designed to ensure users of the Certificate understand what the heating and cooling load limits and features determining the load limits (displayed on page 1 of the Certificate) are for the dwelling.

Predicted onsite renewable energy generation

New information about the home’s predicted onsite renewable energy impact has been added to the Certificate.

This section provides a Whole of Home rating for the home that excludes onsite renewable energy generation. It also includes information about how much energy the home is generating and exporting as well as information about the home’s annual greenhouse gas emissions. This information can be used to see if the home is net zero energy and carbon.

Sample of a house (class 1) and apartment (Class 2 and 4 individual unit) Certificate



Predicted Whole of Home annual impact by appliance

New information on the home’s annual energy use, greenhouse gas emissions and cost by appliance and fuel type has been added to the Certificate. This information allows users to understand what impact each appliance is having on the home’s energy use, cost and greenhouse gas emissions.

Graph Key

A new graph key has been added to help a user with difficulty seeing colour access the information in the pie graphs.

For further information please visit: www.nathers.gov.au



Nationwide House Energy
Rating Scheme

Changes to the 2022 NatHERS Certificates

Certificate Check

An expanded checklist has been added to the Certificate. This checklist allows users to check that the home is designed and then built according to the information used to create the NatHERS ratings. Users can check this at different stages of the home’s design and construction.

Sample of parts of a house (class 1) and apartment (Class 2 and 4 individual unit) Certificate

[#000000000-00] NatHERS Certificate 0.0 Star rating and 00 Whole of Home rating as of [Date]

Certificate check

The checklist covers important items impacting the dwelling’s ratings. It is recommended that the accuracy of the whole certificate is checked.

Note: The boxes indicate when and who should check each item. It is not mandatory to complete this checklist.

	Approval stage		Construction stage		Occupancy/other
	Assessor checked	Consent authority/surveyor checked	Builder checked	Consent authority/surveyor checked	
Genuine certificate check					
Does this Certificate match the one available at the web address or QR code verification link on the front page?		☐	☐	☐	☐
Does the NatHERS certificate number on the NatHERS-stamped plans match the number on this Certificate?		☐	☐	☐	☐

Thermal bridging Schedule

A new thermal bridging schedule for steel framed elements has been added to the Certificate. This schedule includes detailed information about the thermal bridging inputs used to create the NatHERS ratings.

Thermal bridging schedule for steel frame elements

Building element	Steel section dimensions (height x width, mm)	Frame spacing (mm)	Steel thickness (BMT,mm)	Thermal break (yes/no)

Schedules

New appliance, onsite renewable energy and battery schedules have been added to the Certificate.

These schedules include detailed information about the appliances, onsite renewable energy and battery systems that were used to create the Whole of Home performance rating.

[#000000000-00] NatHERS Certificate 0.0 Star rating and 00 Whole of Home rating as of [Date]

Appliance schedule

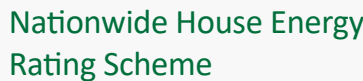
(not applicable if a Whole of Home performance assessment is not conducted for this certificate)

Note: A flat assumption of 5W/m² is used for lighting, therefore lighting is not included in the appliance schedule.

Cooling system

Appliance/ system type	Location	Fuel type	Minimum efficiency/ performance	Recommended capacity
Ducted refrigerative air conditioning (heat pump)	Kitchen/Dining/Living	Electric	00	00
Ducted refrigerative air conditioning (heat pump)	Bedroom 1	Electric	00	00
Ducted refrigerative air conditioning (heat pump)	Bedroom 2	Electric	00	00

For further information please visit: www.nathers.gov.au



Changes to the 2022 NatHERS Certificates

Explanatory Notes

The explanatory notes have been updated to include information about the new features on the Certificate.

Glossary

The glossary has been updated to include information on the new features on the Certificate.

Sample of a house (class 1) and apartment (Class 2 and 4 individual unit) Certificate



[#000000000-00] NatHERS Certificate

0.0 Star rating and 00 Whole of Home rating as of [Date]



Explanatory notes

About this report

NatHERS ratings are a reliable guide for comparing different dwelling designs and to demonstrate that designs meet the energy efficiency requirements in the National Construction Code.

NatHERS ratings use computer modelling to evaluate a home's energy efficiency and performance. They use localised climate data and standard assumptions on how people use their home to predict the heating and cooling energy loads and energy value* of the whole home. The thermal performance star rating uses the home's building specifications, layout, orientation and fabric (i.e. walls, windows, floors, roofs and ceilings) to predict the heating and cooling energy loads.

The Whole of Home performance rating uses information about the home's appliances and on-site energy generation and storage to estimate the home's energy value*.

The actual energy loads, cost and greenhouse gas emissions of a home may vary from that predicted. This is because the assumptions will not always match the actual occupant usage patterns. For example, the number of occupants and how people use their appliances will vary.

Energy efficient homes use less energy, are warmer on cool days, cooler on hot days and cost less to run.

Accredited assessors

For quality assured NatHERS Certificates, always use an accredited or licensed assessor registered with an Assessor Accrediting Organisation (AAO). AAOs have strict quality assurance processes, and professional development requirements ensuring consistently high standards for assessments.

Non-accredited assessors (Raters) have no ongoing training requirements and

are not quality assured.

Any queries about this report should be directed to the assessor. If the assessor is unable to address questions or concerns, contact the AAO specified on the front of this certificate.

Disclaimer

The NatHERS Certificate format is developed by the NatHERS Administrator. However, the content in the certificate is entered by the assessor. It is the assessor's responsibility to use NatHERS accredited software correctly and follow the NatHERS Technical Note. To produce a NatHERS Certificate.

The predicted annual energy load, cost and greenhouse gas emissions in this NatHERS Certificate are an estimate based on an assessment of the dwelling's design by the assessor. It is not a prediction of actual energy use, cost or emissions. The information and ratings may be used to compare how other dwellings are likely to perform when used in a similar way.

Information presented in this report relies on a range of standard assumptions (both embedded in NatHERS accredited software and made by the assessor who prepared this report), including assumptions about occupancy, behaviour, appliance performance, indoor air temperature and local climate.

Not all assumptions made by the assessor using the NatHERS accredited software tool are presented in this report and further details or data files may be obtained from the assessor.

Glossary

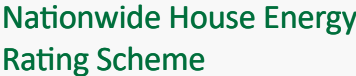
Annual energy load	the predicted amount of energy required for heating and cooling, based on standard occupancy assumptions.
AFRC	Australian Fenestration Rating Council
Assessed floor area	the floor area modelled in the software for the purpose of the NatHERS assessment. Note, this may not be consistent with the floor area in the design documents.
Ceiling penetrations	features that require a penetration to the ceiling, including downlights, vents, exhaust fans, range hoods, chimneys and flues. Excludes fixtures attached to the ceiling with small holes through the ceiling for wiring, e.g. ceiling fans, pendant lights, and heating and cooling ducts.
Conditioned	a zone within a dwelling that is expected to require heating and cooling based on standard occupancy assumptions. In some circumstances it will include garages.
COP	Coefficient of performance
Custom windows	windows listed in NatHERS software that are available on the market in Australia and have a WERS (Window Energy Rating Scheme) rating.
Default windows	windows that are representative of a specific type of window product and whose properties have been derived by statistical methods.
EER	Energy Efficiency Ratio, measure of how much cooling can be achieved by an air conditioner for a single kWh of electricity input
Energy use	This is your homes rating without solar or batteries.
Energy value	The net cost to society including, but not limited to, costs to the building user, the environment and energy networks (as defined in the ABCB Housing Provisions Standard).
Entrance door	these signify ventilation benefits in the modelling software and must not be modelled as a door when opening to a minimally ventilated corridor in a Class 2 building.
Exposure	see exposure categories below.
Exposure category – exposed	terrain with no obstructions e.g. flat grazing land, ocean-frontage, desert, exposed high-rise unit (usually above 10 floors).
Exposure category – open	terrain with few obstructions at a similar height e.g. grasslands with few well scattered obstructions below 10m, farmland with scattered sheds, lightly vegetated bush blocks, elevated units (e.g. above 3 floors).
Exposure category – suburban	terrain with numerous, closely spaced obstructions below 10m e.g. suburban housing, heavily vegetated bushland areas.
Exposure category – protected	terrain with numerous, closely spaced obstructions over 10 m e.g. city and industrial areas.
Horizontal shading feature	provides shading to the building in the horizontal plane, e.g. eaves, verandahs, pergolas, carports, or overhangs or balconies from upper levels.
National Construction Code (NCC) Class	the NCC groups buildings by their function and use, and assigns a classification code. NatHERS software models NCC Class 1, 2 or 4 buildings and attached Class 10a buildings. Definitions can be found at www.abcb.gov.au .
Net zero home	a home that achieves a net zero energy value*
Opening percentage	the operability percentage or operable (movable) area of doors or windows that is used in ventilation calculations.
Provisional value	an assumed value that does not represent an actual value. For example, if the wall colour is unspecified in the documentation, a provisional value of 'medium' must be modelled. Acceptable provisional values are outlined in the NatHERS Technical Note and can be found at www.nathers.gov.au
Recommended capacity	this is the capacity or size of equipment that is recommended by NatHERS to achieve the desired comfort conditions in the zone or zones serviced. This is a recommendation and the final selection sizing should be confirmed by a suitably qualified person.
Reflective wrap (also known as foil)	can be applied to walls, roofs and ceilings. When combined with an appropriate air gap and emissivity value, it provides insulative properties.
Roof window	for NatHERS this is typically an operable window (i.e. can be opened), will have a plaster or similar light well if there is an attic space, and generally does not have a diffuser.
Shading features	includes neighbouring buildings, fences, and wing walls, but excludes eaves.
Solar heat gain coefficient (SHGC)	the fraction of incident solar radiation admitted through a window, both directly transmitted as well as absorbed and subsequently released inward. SHGC is expressed as a number between 0 and 1. The lower a window's SHGC, the less solar heat it transmits.
Skylight (also known as roof lights)	for NatHERS this is typically a moulded unit with flexible reflective tubing (light well) and a diffuser at ceiling level.
STCs	Small-scale Technology Certificates, certificates created by the REC registry for renewable energy technologies that may be bought and sold as part of the Small-scale Renewable Energy Scheme operated by the Clean Energy Regulatory
Thermal breaks	are materials with an R-value greater than or equal to 0.2 that must separate the metal frame from the cladding. This includes, but is not limited to, materials such as timber battens greater than or equal to 20mm thick, continuous thermal breaks such as polystyrene insulation sheathing, plastic strips or furring channels.
U-value	the rate of heat transfer through a window. The lower the U-value, the better the insulating ability.
Unconditioned	a zone within a dwelling that is assumed to not require heating and cooling based on standard occupancy assumptions.
Vertical shading features	provides shading to the building in the vertical plane and can be parallel or perpendicular to the subject wall/window. Includes privacy screens, other walls in the building (wing walls), fences, other buildings, vegetation (protected or self heritage trees).
Window shading device	a device fixed to buildings that provides shading e.g. window awnings or screens but excludes horizontal or vertical shading features* (eg eaves and balconies)

* Refer to glossary.

Generated on [date] using [software] for [address]

Page 10 of 10

For further information please visit: www.nathers.gov.au



Changes to the 2022 NatHERS Certificates

Sample of a unit block (class 2) summary Certificate

Class 2 Summary Certificate

This Certificate provides information about sole-occupancy units of a Class 2 building.

**Nationwide House Energy Rating Scheme®
Class 2 Summary
NatHERS® Certificate No. [#000000000-00]**

Generated on [date] using [software and version]
[other boilerplate text other boilerplate text other boilerplate text other
boilerplate text other boilerplate text other boilerplate text]

Property

Address	[00 Street, Suburb, State/Territory, Postcode]
Lot/DP	[number]
NatHERS Climate Zone	[number]



 Accredited assessor

Name [assessor name]
Business name [business name]
Email [email address]
Phone [00 0000 0000]
Accreditation No. [0000 000 000]

Assessor Accrediting Organisation
[name of Assessor Accrediting Organisation]

Verification

To verify this certificate, scan the QR code or visit [\[Hstar-dev.azurewebsites.net/QR/Generate?p=MalcPjqJ.\]](https://[Hstar-dev.azurewebsites.net/QR/Generate?p=MalcPjqJ.])

When using either link, ensure you are visiting hstar-dev.azurewebsites.net

A square QR code located in the bottom right corner of the slide, used for verifying the certificate.

National Construction Code (NCC) requirements

The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J3D3 and J3D15 of NCC Volume One.

NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.

The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au.

Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.

Summary of all dwellings

Certificate number and link	Unit Number	Heating load (MJ/m2/p.a.)	Cooling load (MJ/m2/p.a.)	Total load (MJ/m2/p.a.)	Star Rating	Whole of Home Rating
Modelled average		0000.0	0000.0	0000.0	0.0	n/a
000000000000	A1	0000.0	0000.0	0000.0	0.0	000
000000000000	A2	0000.0	0000.0	0000.0	0.0	000

* Refer to glossary.
Generated on [date] using [software] for [address]

Page 1 of

Thermal performance Star rating



**NATIONWIDE
HOUSE**
ENERGY RATING SCHEME

The rating above is the average of all dwellings in this summary

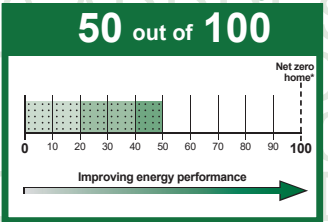
For more information on
your dwelling's rating see:
www.nathers.gov.au

NCC heating and cooling maximum loads MJ/m²/p.a.

Limits taken from ABCB Standard 2022:

	Heating	Cooling
Modelled block average	0000.0	0000.0
Maximum allowable limit	0000.0	0000.0

Whole of Home performance rating



The rating above is the lowest of all the dwellings in this summary

Thermal performance star rating

The new NatHERS Class 2 Summary Certificate retains the average thermal performance star rating to demonstrate compliance with the heating and cooling loads for sole-occupancy units of a Class 2 building in NCC 2022.

NCC heating and cooling maximum loads

New information about the Heating and Cooling Load Limits has been added to the Summary Certificate. This section shows the average heating and cooling loads limits for the unit block. This information will allow users of the Certificate, such as certifiers, to quickly assess whether the unit block load limits have been met, while also allowing them to consider local variations.

Whole of Home Performance Rating

The new Whole of Home performance rating in the Class 2 summary Certificate represents the lowest individual rating for a unit in the unit block. The rating allows users of the Certificate, such as certifiers, to quickly assess whether the dwelling meets the energy usage requirement in NCC 2022. Note that there is no average rating requirement for Whole of Home.

For further information please visit: www.nathers.gov.au

Sample of a unit block (class 2) summary Certificate



Nationwide House Energy Rating Scheme

Changes to the 2022 NatHERS Certificates

Summary of all dwellings

The summary table continues onto the following pages to account for all the units in the unit block.

Explanatory Notes

Updated explanatory notes include information about the new features on the Summary Certificate.

[#000000000-00] NatHERS Certificate 0.0 Star Rating and 00 Whole of Home Rating as of [Date]



Summary of all dwellings (continued)

Certificate number and link	Unit Number	Heating load (MJ/m ² /p.a.)	Cooling load (MJ/m ² /p.a.)	Total load (MJ/m ² /p.a.)	Star Rating	Whole of Home Rating
0000000000	A1	0000.0	0000.0	0000.0	0.0	000
0000000000	A2	0000.0	0000.0	0000.0	0.0	000
0000000000	A3	0000.0	0000.0	0000.0	0.0	000
0000000000	A4	0000.0	0000.0	0000.0	0.0	000
0000000000	A5	0000.0	0000.0	0000.0	0.0	000
0000000000	A6	0000.0	0000.0	0000.0	0.0	000
0000000000	A7	0000.0	0000.0	0000.0	0.0	000
0000000000	A8	0000.0	0000.0	0000.0	0.0	000
0000000000	A9	0000.0	0000.0	0000.0	0.0	000
0000000000	A10	0000.0	0000.0	0000.0	0.0	000
0000000000	A11	0000.0	0000.0	0000.0	0.0	000
0000000000	A12	0000.0	0000.0	0000.0	0.0	000

Explanatory notes

About the ratings

The thermal performance star rating in this Certificate is the average rating of all NCC Class 2 dwellings in an apartment block. The Whole of Home performance rating in this Certificate is the lowest rating for the apartment block. Individual unit ratings are listed in the 'Summary of all dwellings' section of this Certificate.

NatHERS ratings use computer modelling to evaluate a home's energy efficiency and performance. They use localised climate data and standard assumptions on how people use their home to predict the energy loads and societal cost. The thermal performance star rating uses the home's building specifications, layout, orientation and fabric (i.e. walls, windows, floors, roofs and ceilings) to predict the heating and cooling energy loads. The Whole of Home performance rating uses information about the home's appliances and onsite energy production and storage to estimate the homes societal cost.

For more details about an individual dwelling's assessment, refer to the individual dwelling's NatHERS Certificate (accessible via link).

Accredited Assessors

For high quality NatHERS Certificates, always use an accredited or licenced assessor registered with an Assessor Accrediting Organisation (AAO). AAOs have strict quality assurance processes, and professional development requirements ensuring consistently high standards for assessments.

Non-accredited assessors (Raters) have no ongoing training requirements and are not quality assured.

Licensed assessors in the Australian Capital Territory (ACT) can produce assessments for regulatory purposes only, using endorsed software, as listed on the ACT licensing register.

Any queries about this report should be directed to the assessor. If the assessor is unable to address questions or concerns, contact the AAO specified on the front of this certificate.

Disclaimer

The NatHERS Certificate format is developed by the NatHERS Administrator. However, the content in certificates is entered by the assessor. It is the assessor's responsibility to use NatHERS accredited software correctly and follow the NatHERS Technical Note to produce a NatHERS Certificate.

The predicted annual energy use, cost and greenhouse gas emissions in this NatHERS Certificate are an estimate based on an assessment of the dwelling's design by the assessor. It is not a prediction of actual energy use, cost or emissions. The information and ratings may be used to compare how other dwellings are likely to perform when used in a similar way.

Information presented in this report relies on a range of standard assumptions (both embedded in NatHERS accredited software and made by the assessor who prepared this report), including assumptions about occupancy, behaviour, appliance performance, indoor air temperature and local climate.

Not all assumptions made by the assessor while using the NatHERS accredited software tool are presented in this report and further details or data files may be available from the assessor.

* Refer to glossary.
Generated on [date] using [software] for [address]

Page 2 of 2

For further information please visit: www.nathers.gov.au



Nationwide House Energy
Rating Scheme

Changes to the 2022 NatHERS Certificates

Non-accredited Report

A non-accredited report is generated by a non-accredited assessor (or rater) using NatHERS accredited software.

A non-accredited assessor (rater) is not required to have any formal qualifications, insurance, ongoing professional development or quality assurance checks on their ratings.

To differentiate between this report and an accredited certificate, it will be produced in black and white and have no NatHERS logos.

It will still have much of the same information to help users evaluate a home's thermal performance and energy efficiency.

Thermal Performance Star Rating

The non-accredited report will not show the thermal performance star rating graphic. The star rating will be written as text.

Sample of a non-accredited Report

Residential energy rating report Non-accredited No. #000000000-00

Generated on [date] using [software and version]

This report was created using NatHERS accredited software but the non-accredited assessor (rater) is not accredited under NatHERS and this report is not accredited as being compliant with NatHERS.
Reliance on this report is accordingly at your own risk.

Property		Thermal performance Star rating	
Address	[00 Street, Suburb, State/Territory, Postcode]	X.X star rating [XX.X] MJ/m ² Predicted annual energy load for heating and cooling based on standard occupancy assumptions.	
Lot/DP	[number]		
NCC class*	[number]		
Floor/all Floors Type	[dwelling entrance floor] of [total no. of floors] floors [new/renovation/existing]		
Plans			
Main plan	[plan number, version & date]		
Prepared by	[name of preparer of plans]		
Construction and environment		Thermal performance (MJ/m²)	
Assessed floor area (m ²)*		Limits taken from ABCB Standard 2022.1	
Conditioned*	000.0	Heating	Cooling
Unconditioned*	0.0	0000.0	0000.0
Total	0.0	Modelling	0000.0
Garage	0.0	Load limits	0000.0
Exposure type	[exposure]	Features determining load limits	
NatHERS climate zone	[number, town/suburb]	Floor type (lowest conditioned area) [Type]	
		NCC climate zone 1 or 2 [Y/N/NA]	
		Outdoor living area [Y/N/NA]	
		Outdoor living area ceiling fan [Y/N/NA]	
Rater**		Whole of Home performance rating	
Name	[assessor name]	50 out of 100	
Business name	[business name]		
Email	[email address]		
Phone	[00 0000 0000]		
Declaration of interest	[declaration]		
NCC Requirements			
BCA provisions	[Volume 1/Volume 2]		
State/Territory variation	[Yes/No]		
National Construction Code (NCC) requirements			
The NCC allows the use of NatHERS accredited software to comply with the energy efficiency requirements for houses (Class 1 buildings) and apartments (Class 2 sole-occupancy units and Class 4 parts of buildings). The applicable requirements for houses are detailed in Specification 42 of NCC Volume Two. For apartments the requirements are detailed in clauses J3D3 and J3D15 of NCC Volume One.			
NCC 2022 includes enhanced thermal performance requirements for houses and apartments. It also includes a new whole-of-home annual energy use budget which applies to the major equipment in the home.			
The NCC, and associated ABCB Standards and support material, can be accessed at www.abcb.gov.au .			
Note, variations and additions to the NCC energy efficiency requirements may apply in some states and territories.			
* Refer to glossary. ** Refer explanatory notes. Generated on [date] using [software] for [address]			
		Verification	
		To verify this report, scan the QR code or visit [Hstar-dev.azurewebsites.net/QR/Generate?p=MialcP]qJ.] When using either link, ensure you are visiting hstar-dev.azurewebsites.net	
		Page 1 of 10	

For further information please visit: www.nathers.gov.au

Thermal Performance

New information about the heating and cooling load limits and features determining the load limits has been added and will be displayed on the non-accredited report also. This information will allow users of the report, such as certifiers, to quickly assess whether the NCC load limits have been met, while also allowing them to consider local variations. Additional explanatory information about the limits and settings is provided on page 2 of the report.

Whole of Home performance rating

For the non-accredited report and similar to the thermal performance star rating, the Whole of Home rating will not show the bar rating graphic. The rating will be written in text.

It will still provide information to show how the home meets or beats the new NCC annual energy use budget.

This rating along with the additional information provided on page 2 of the Certificate can be used to indicate if the home is 'net zero energy' or 'net zero carbon.'



Nationwide House Energy
Rating Scheme

Changes to the 2022
NatHERS Certificates

About the ratings

New explanatory information is provided about each of the NatHERS ratings. This information is designed to ensure users of the report understand what each rating is measuring.

Heating and Cooling Load Limits

New explanatory information is provided about the heating and cooling load limits.

This information is designed to ensure users of the report understand what the heating and cooling load limits and features determining the load limits (displayed on page 1 of the report) are for the dwelling.

Predicted onsite renewable energy generation

New information about the home’s predicted onsite renewable energy impact has been added to the non-accredited report also.

This section provides a Whole of Home rating for the home that excludes onsite renewable energy generation. It also includes information about how much energy the home is generating and exporting as well as information about the home’s annual greenhouse gas emissions. This information can be used to see if the home is net zero energy and carbon.

Sample of a non-accredited Report

[#000000000-00] Non-accredited report

About the ratings

Thermal performance rating

NatHERS thermal software models the expected heating and cooling energy loads using information about the design, construction, climate and common patterns of household use. The thermal performance rating (shown as a written rating on this Report) does not take into account appliances, apart from the airflow impacts from ceiling fans.

Whole of Home performance rating

NatHERS Whole of Home software uses the heating and cooling energy loads combined with the energy performance of the home's appliances (heating, cooling, hot water, lighting, pool/spa pump and onsite renewable energy generation and storage) and models the expected energy value* of the whole home. The Whole of Home performance rating is shown as a score out of 100 on this Report.

Heating & Cooling Load Limits

Additional information

In some locations under the NCC NatHERS pathway, separate heating and cooling load limits may apply. Minimum required star ratings in northern parts of Australia may also be affected by the presence or absence of an outdoor living area and/or an outdoor living area ceiling fan. Refer to the *ABCB NatHERS heating and cooling load limits Standard 2022.1* for details or contact the relevant local building regulating authority, noting that State and Territory variations may also apply.

Setting options:

Floor type:
CSOG – Concrete Slab on Ground
SF – Suspended Floor (or a mixture of CSOG and SF)
NA – Not Applicable

NCC climate Zone 1 or 2:
Yes
No
NA – not applicable

Outdoor living area:
Yes
No
NA – not applicable

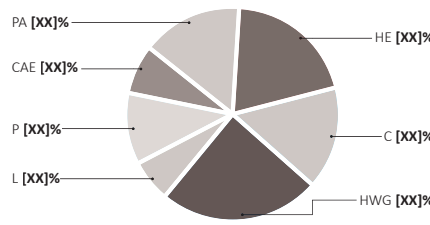
Outdoor living area ceiling fan:
Yes
No
NA – not applicable

0.0 Star rating and 00 Whole of Home rating as of [Date]

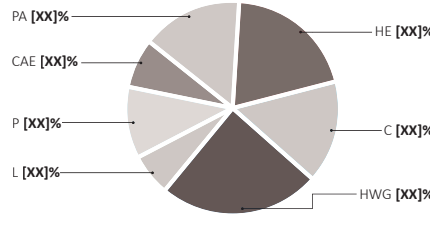
Predicted Whole of Home annual impact by appliance

Shows the contribution each appliance has on the home's annual energy use, greenhouse gas emissions and cost without solar.

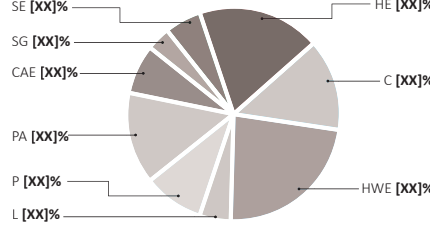
Energy use:



Greenhouse gas emissions:



Cost:



Graph Key:

Colour:	Code:	Name:	Fuel type:
[Dark Brown]	HE	Heating	electric
[Medium Brown]	HG	Heating	gas
[Light Brown]	HW	Heating	wood
[Lightest Brown]	C	Cooling	electric
[Dark Brown]	HWE	Hot water	electric
[Medium Brown]	HWG	Hot water	gas
[Light Brown]	L	Lights	electric
[Lightest Brown]	P	Pool/Spa equipment	electric
[Dark Brown]	PA	Plug-in appliances	electric
[Medium Brown]	CAE	Cooking appliances	electric
[Light Brown]	CAG	Cooking appliances	gas
[Lightest Brown]	SG	Supply charge	gas
[Dark Brown]	SE	Supply charge	electric

Predicted onsite renewable energy impact

Your Whole of Home energy use* rating excluding onsite renewable energy generation is [00] out of 100.

This home's annual greenhouse gas emissions: [0000]kg CO2e (with solar)
[0000]kg CO2e (without solar)

Predicted annual electricity use: [0000] kWh

Exported to the grid: [00]%

Used by the home: [00]%

* Refer to glossary. ** Refer explanatory notes.
Generated on [date] using [software] for [address]

Page 2 of 10

Predicted Whole of Home annual impact by appliance

New information on the home’s annual energy use, greenhouse gas emissions and cost by appliance and fuel type will be included on the non-accredited report as well. This information allows users to understand what impact each appliance is having on the home’s energy use, cost and greenhouse gas emissions.

Graph Key

A new graph key has been added to help a user with difficulty seeing colour access the information in the pie graphs.

For further information please visit: www.nathers.gov.au

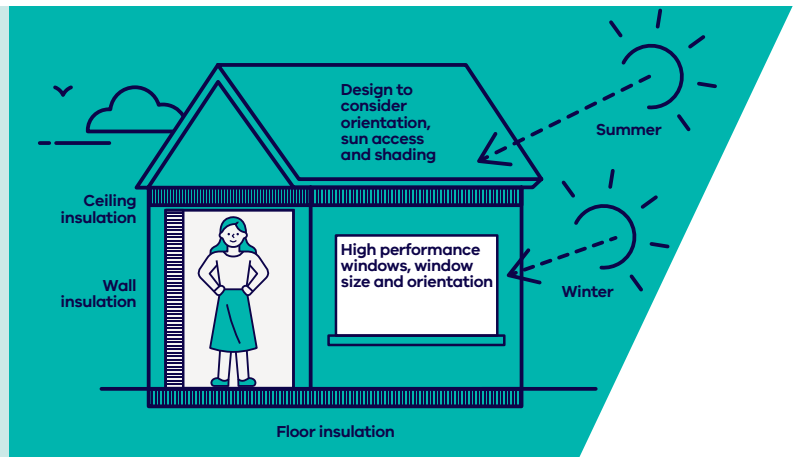
Fact sheet for homebuyers

Energy efficient new homes

Ways to meet new efficiency standards that will come into effect from October 2023

1. Design and Build your home to a minimum thermal performance rating of 7 stars so it needs less energy to heat and cool

Use less energy



2. Choose fixed appliances that are more energy efficient so you need less energy to power your home

Use efficient appliances



3. Install rooftop solar to help offset your remaining energy usage and meet your Whole-of-Home budget

Use renewable energy



Consider other opportunities to future-proof your home and better manage your energy usage. You could:

- Build an all-electric home to maximise benefits of your solar
- Install batteries
- Optimise energy use through energy monitoring and timers
- Use more efficient plug-in appliances
- Select at least 4 star WELS hot water fixtures, eg showers and washing machines



Meeting the Whole-of-Home annual energy use budget

Hot water, heating, cooling, and lighting account for **70-80% of an average home's energy use** so these services provide ample scope for energy savings. The new standards will now require new homes to meet a Whole-of-Home annual energy usage budget to encourage the installation of efficient fixed appliances.

This Whole-of-Home approach sets a tailored budget for new homes to be met through a flexible combination of:

- type and efficiency rating of specific fixed appliances (hot water, heating, cooling, lighting and pool and spa pumps)
- installation of rooftop solar to offset any energy use above the budget cap

Home builders can choose the fuel type (gas or electricity), technology and efficiency rating of fixed appliances. This is as long as they meet the annual energy usage budget or install enough rooftop solar to offset the above-budget energy usage.

Key benefits



Lower energy bills

National energy efficiency standards for new homes have not been updated for more than a decade, during which time gas and electricity prices have risen dramatically. With prices expected to keep rising, improving efficiency to reduce the amount of energy needed to keep your home comfortable, is an obvious way to cut energy costs.

Research has shown cost savings from \$300 to over \$1,000 a year are achievable through energy efficient homes.*



More comfortable homes

Energy efficient homes are more comfortable to live in year-round, helping to protect people from the health risks of chronic cold and extreme weather events like heatwaves.



Lower greenhouse emissions

Victoria has set a target of net-zero emissions by 2050 with ambitious but achievable targets to reduce the state's greenhouse gas emissions from 2005 levels by 45–50 per cent by 2030.

Residential buildings are responsible for 11% of Australia's greenhouse gas emissions and offer many of the lowest cost opportunities for cutting emissions across the economy.



A greener energy grid

Making homes more energy efficient, helps to reduce demand on the electricity system at peak times, reducing the need for expensive network infrastructure upgrades that contribute to higher prices. Reducing demand will also reduce the costs as the energy system transitions to renewable energy.

Costs impacts for homebuyers

If smart design is considered early in the process of a new home build, achieving 7 stars can be met without significant additional costs or changes in materials or construction, and will deliver savings on energy bills. Any additional upfront costs of construction required to meet the new 7 star standards are expected to be minimal, particularly in the context of energy cost savings made over the multi-decade lifetime of the home and an average 30-year mortgage.

To help reduce the cost impacts for consumers the Victorian Government is funding rebates under the [Solar Homes Program](#).

*Analysis undertaken for NCC 2022, showed the new efficiency standards would save an average Victorian household at least \$300 a year in energy bills. More recent analysis by [Renew](#) found even higher savings – \$1,056 for an all-electric home in Melbourne and slightly less for a dual fuel home.



Henley 2022 Good Friday Appeal Charity home. Double storey Vogue 34 design.

ABOUT THIS HOME

NatHERS Rating: 7.1 Stars

Size: 257.3m² – 4 bedrooms, 3 bathrooms, 4 living spaces

Whole-of-Home: Ducted reverse cycle heating and cooling, heat pump hot water, induction cooking, 6kW solar PV, 13.5kWh battery

Annual Energy Bill: \$591

Looking ahead

The new energy efficiency provisions under NCC 2022 are available for use from 1 October 2022, with a 12-month transition period to allow the construction industry to make necessary adjustments and bring new projects in line with the uplifted standards. During the 12-month transition, homes will be permitted to be built to either NCC 2019 or the new improved NCC 2022 standards.

The new standards will be mandatory from 1 October 2023.

For more information on the new standards or how to future-proof your home with efficiency improvements, please visit energy.vic.gov.au/7starhomes and <https://www.sustainability.vic.gov.au/energy-efficiency-and-reducing-emissions/building-or-renovating>.

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4

Increase zoning:

By including more internal doors in the design, mechanical heating and cooling estimates in NatHERS can be applied more accurately. Places to include doors may be at the end of corridors, the bottom of stairs or to close off occasionally used rooms, like media and rumpus rooms.

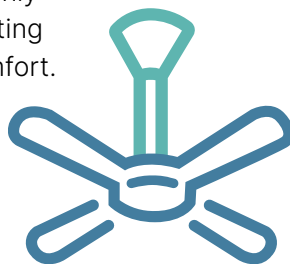


5

Insulate more: Depending on the climate, ratings may be improved by increasing the R value of wall, floor and ceiling insulation or extending insulation to new areas e.g. slab edge, underneath the slab, the garage wall and internal walls (especially to unconditioned zones, like laundries and bathrooms). Care must be taken to not increase condensation risk. For more information, see www.yourhome.gov.au/passive-design/insulation, www.yourhome.gov.au/passive-design/condensation and www.abcb.gov.au/sites/default/files/resources/2022/Handbook-condensation-in-buildings-2019.pdf

6

Plan for fans: Specifying ceiling fans in all rooms will not only improve your NatHERS rating but the homeowners' comfort.



7

Check your colours: Improvements to a NatHERS rating can also be achieved by adjusting the colour of the roof and the external walls. For example by moving away from using darker roof colours in warmer climates. Some software can even adjust window frame colours which can make a surprising difference. Talk to your clients about their choices early.

8

Look at lighting: NatHERS rewards homes with fewer ceiling penetrations with higher ratings. Consider whether your design can delete downlights that aren't IC4 rated, roof windows and skylights and replace these with other lighting options that allow for continuous ceiling insulation.

**Find out more**

Your Home provides free house plans and case studies that meet 7 stars available at www.YourHome.gov.au

